WEB ANALYTICS · FINAL PROJECT PITCH

Smart Supermarket Basket Optimizer

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Goal and motivation

THE QUESTION

Where, and in how many stores, should I shop to pay the least for my shopping list?

Everyday spending
Groceries are a weekly cost for everyone, so small savings add up.

Hard to compare by hand
Every chain has its own website, pack sizes and units.

Rich public data
Online prices can be collected automatically and analyzed.

Problem Formulation - TRAVELING PURCHASER PROBLEM

A buyer leaves home with a list and chooses which stores to visit, and in what order, so that purchase cost + travel cost is as low as possible.

$$\min_S \ \Sigma_i q_i \cdot \min_{s \in S} p_{i,s} \ + \ \lambda \cdot L(S) \ + \ \mu \cdot (|S| - 1)$$

λ: € per km of route, L(S) · μ: € per extra stop · S: stores visited

TPP	Our project
Depot	User's home (postal code). Nominatim
Markets	3 nearest stores per chain, up to 3 stops. OSM / Overpass
Product list	User's basket
Price at each market	Scraped or API price of the matched product in the given chain
Edge costs	Straight-line km × 1.3 (Haversine)

TPP in Practice - COMPLEXITY AND VALUE

Why it is hard

- **NP-hard:** if each item is sold in only one store, the TPP becomes the Travelling Salesman Problem
- **Coupled choices:** a cheaper store can lose its saving on the detour, so stores and route must be chosen together

$\sim 10^{20}$	**Without limits:** ordered routes over any subset of 21 stores
1,155	**Our limits:** store sets with 3 nearest per chain, ≤ 1 per chain, ≤ 3 stops
6,069	**What we score:** routes at most, so trying all of them is exact and instant

USEFULNESS

The exact plan costs **26.11 €** against **28.10 €** when each item is bought at its cheapest chain: 7% less, better in 29 of 30 baskets. One store was enough in 17 of 30.

30 random 5-item baskets near UC3M Leganés, stores within 3 km, 0.25 €/km, 1 € per extra stop. Prices are provisional.

Data Sources and Why

Source	How	Why this source?
DIA, Ahorramás, Alcampo, Aldi	Web scraping (Selenium, BeautifulSoup)	Prices are only available on their websites
Mercadona, BM, Consum	Web APIs (JSON endpoints)	Structured data, faster and more stable than scraping
Carrefour	Dropped out because of Cloudflare blockage	Major chain, but its anti-bot protection makes collection hard
OpenStreetMap	Nominatim and Overpass APIs	Free geocoding and supermarket locations

Pipeline: raw CSV per chain, then cleaning and unit normalization, then one processed dataset.

Similarity Search - WHY, TECHNIQUES AND PLAN

5 of 7 chains publish no barcode, and names differ: “Yogur líquido con fresa y plátano 6x100 ml” at BM is “Actimel Kids 6+2 gratis Sabor Fresa-Plátano” at Consum.

Techniques so far

- **Cleaning:** names, brand and size
- **TF-IDF** on character 3-5-grams, cosine
- **Product types:** 69 types, 56 anchors
- **Ground truth:** 681 barcodes, 97 labels

TF-IDF baseline: finding each of the 681 BM products among 2,953 Consum products

Text used	Top-1	Top-5
Name only	41%	77%
Name + brand	73%	95%

PLAN

1. Compare sentence embeddings with TF-IDF on the same pairs

2. Tune the thresholds τ and δ on the hand labels

3. Extend to all 13,253 products and held-out categories

Where we use it: same-product basket prices, chain comparison and free-text search.

Planned Analysis - STATUS

Done

Normalize

Prices converted to €/kg, €/L or €/unit.

Optimize the basket

Best set of up to 3 stores, minimizing product cost plus travel cost.

Visualize

Streamlit dashboard with basket costs, a map of nearby stores and the recommended plan.

In progress

Match

Equivalent products across chains with text similarity.

Compare chains

Price dispersion by product, a price index per chain and outlier detection.

Next

Automated data updates

Weekly scraping and API requests to keep product catalogs and prices up to date.

Price changes

Track how prices move between weekly snapshots.

This is a plan: techniques may change as the data grows.

State of the Art

Feature	SoySuper	FindIt	Our Project
Full basket comparison	✅	✅	✅
Product equivalents	✅	Not specified	Planned
Shopping approach	Online delivery	Price comparison	Physical stores
Multiple-store basket optimization	Not offered	Not documented	✅ Up to 3 stores
Travel cost optimization	Not offered	Not documented	✅ TPP-based

Difficulties - AND HOW WE ARE ADDRESSING THEM

Difficulty	What we are doing
Carrefour blocked by Cloudflare; DIA API returns 403	Dropped Carrefour; kept a Selenium scraper for DIA
Selenium scrapers are fragile and partial	Using JSON APIs where they exist: Mercadona, BM, Consum
Each chain writes prices differently	One adapter per chain; quality flags instead of deleting rows
Classifier errors: pomace oil among extra-virgin oils	Adding negative anchors and evaluating on 97 hand labels
Overpass and Nominatim can fail during a demo	Falling back to a saved list of 860 Madrid stores
robots.txt disallows the APIs we use	Academic use, disclosed: 1 s pause, identifying User-Agent

Live Demo so far